direction of the current is same. Thus, the frequency of rectified output voltage is two times that of the input frequency.

The input and output waveforms are shown in Figure 4.11(b).

$$Output\ of\ the\ full\ wave\ rectifier = \frac{2E_{max}}{\pi}$$

$$\therefore\ The\ Average\ DC\ Voltage = 0.637\,E_{max}$$

$$Ripple\ Factors = 2 \times f$$

$$Efficiency = 81.2\%$$

4.6 FULL-WAVE BRIDGE RECTIFIER

As the centre-tapped transformer is expensive and is difficult to implement, bridge rectifier was developed. In this, four diodes are arranged in the form of a bridge-configuration to produce the desired output. Figure 4.12 shows the full wave bridge rectifier circuit.

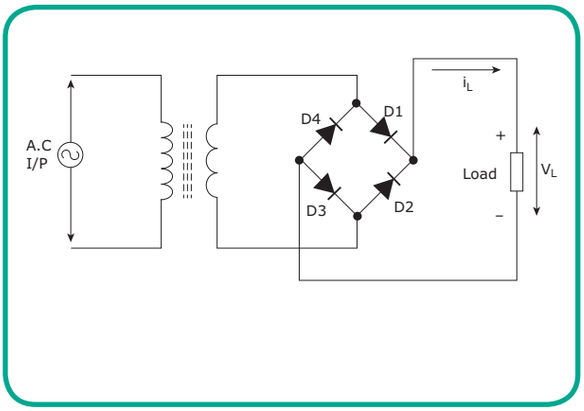


Figure 4.12 Full-wave bridge rectifier circuit

The four diodes labeled D1, D2, D3 and D4 are arranged in series pairs with only two diodes conducting current during each half-cycle. During the positive half cycle of the supply, diodes D1 and D3 conduct in series while D2 and D4 are reverse-biased. The current flows through the load is shown in Figure 4.13.

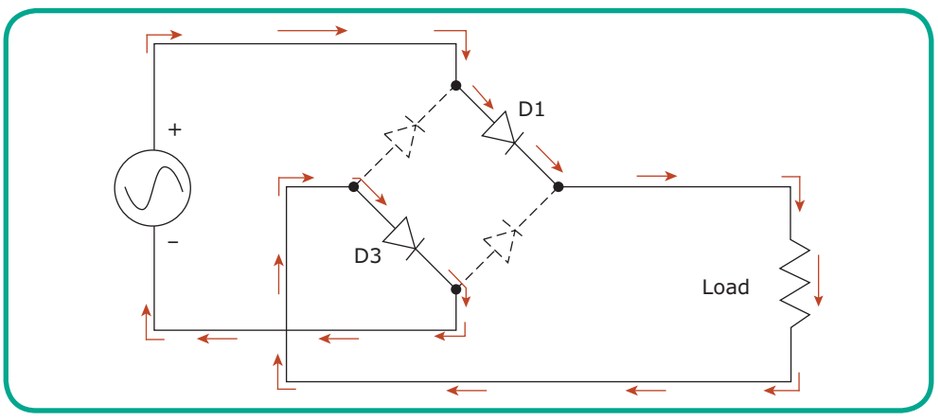


Figure 4.13 Current flows through the load

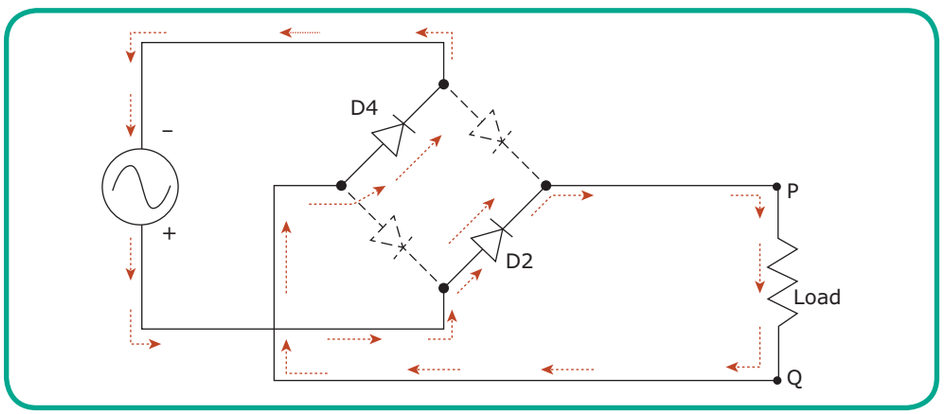


Figure 4.14 Current flows through the load

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